

Native Delegation as a Runtime Primitive

Ligate Labs

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**Native Delegation as a Runtime Primitive: Hot-Key / Master-Key Separation
for Attestation-Native Chains**

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Abstract

Smart-contract wallets (ERC-4337, SafeWallet) and authorization modules (Cosmos authz, Solana fee-payer) implement hot-key / master-key separation as an application-layer or module-layer pattern. The contract or module mediates which key can sign what, when, for how long, and what happens when the key misbehaves. This is the standard pattern on chains with general-purpose smart contracts.

Ligate Chain does not have general-purpose smart contracts. Runtime primitives are how we express anything that elsewhere would be a contract. This paper specifies **native delegation** as a runtime primitive: a delegation transaction type, schema-scoped and action-scoped grants, time-bounds with explicit revocation, and slashing-inheritance rules tied to PoUA reputation evolution. The mechanism is the foundation for the Iris MCP relayer, where autonomous agents act on behalf of a user without holding the user’s master key, and for any future product whose UX is “the user signed once, the agent can act on their behalf for the next T seconds.”

[v0.2 will fill in: the formal delegation tx schema, the slashing-inheritance theorem, the cross-schema-arbitrage cost-to-attack relationship, the comparison table, and the limitations.]

1. Introduction

1.1 The Agent-on-Behalf-of-User Thesis

[v0.2: Why the next surface for blockchain UX is autonomous agents acting on behalf of users, not users themselves. Cite ChatGPT plugins, Anthropic’s Claude tools, Auto-GPT lineage, current restaking-AVS infrastructure. Frame: the chain primitive that needs to land is “the user signed once, the agent can sign on their behalf for the next T seconds, scoped to actions A , with consequences C if the agent misbehaves.”]

1.2 Why Now

[v0.2: Iris is Ligate’s MCP relayer for autonomous agents (v0.5 product target). Cannot ship safely without a delegation primitive: an agent holding the user’s master key is unacceptable; an agent paying gas with no scope-bounds is also unacceptable. The convergence of (1) AI agents reaching production, (2) account-abstraction patterns proven on Ethereum, (3) PoUA reputation accounting at the protocol layer, makes native delegation the right design choice now.]

1.3 The Contract-vs-Runtime Tradeoff

[v0.2: ERC-4337 makes account abstraction work on Ethereum by adding a contract layer above the EVM. Cosmos authz does it with a module. Both work because their host chains run general-purpose VM-style execution. Ligate runs a Sovereign SDK rollup on Celestia; we have runtime primitives, not contracts. Anything that would be an ERC-4337 EntryPoint contract in Ethereum-land is, in Ligate-land, a protocol message type. This is a design choice, not a constraint: protocol-native delegation has cleaner integration with PoUA reputation accounting and slashing.]

1.4 The Central Question

[v0.2: What is the smallest runtime delegation primitive that supports agent-on-behalf-of-user UX, integrates cleanly with PoUA reputation slashing, and enforces scope and

time-bounds without an EVM-like contract layer?]

1.5 Approach in Brief

[v0.2: Three-sentence preview. A `MsgDelegate` transaction type carries master-key signature plus a grant object (scope, time-bounds, slash-inheritance rule). A `MsgRevokeDelegate` transaction nullifies the grant, with an optional grace period. PoUA reputation updates apply to either the master, the hot key, or both, depending on the slash-inheritance rule chosen at grant time.]

1.6 Contributions

[v0.2: Mechanism specification, slashing-inheritance theorem, formal comparison with ERC-4337 / SafeWallet / Cosmos authz / Solana fee-payer, security analysis under five threat models, Iris-specific use case derivations.]

1.6.1 Status of Claims [v0.2: Same panel as PoUA v0.7 §1.6.1: proven, bounded-under-stated-assumptions, empirical-or-heuristic. The slashing-inheritance theorem is a candidate for “proven”; the security claims under colluding-agent assumptions are likely “bounded-under-stated-assumptions”; cross-product UX claims are “empirical-or-heuristic.”]

1.7 Scope and Non-Goals

[v0.2: In scope: hot-key / master-key delegation as a runtime primitive, schema-scoped and action-scoped grants, time-bounds, revocation, slashing inheritance, Iris integration. Out of scope: cross-chain delegation (separate paper), recursive multi-level delegation (deferred to v0.3), hardware-wallet integration semantics (product layer, not protocol), generic account abstraction (we are not building an ERC-4337 equivalent on Ligate).]

1.8 Document Structure

[v0.2: TOC walkthrough.]

2. Background and Related Work

2.1 ERC-4337 and Smart-Contract Wallets

[v0.2: EntryPoint contract architecture. UserOperation pseudo-transaction format. Bundlers and paymasters. Account abstraction goals (sign with anything, sponsor gas, batch operations). Tradeoffs: contract-layer cost, censorship surface, cross-DeFi composability advantages.]

2.2 SafeWallet (formerly Gnosis Safe)

[v0.2: Multisig contracts with K-of-N approval. Module system for custom signing rules. No native scope-grants; modules implement scope-bounded signing as application logic.]

2.3 Cosmos authz Module

[v0.2: Generic delegation module (`x/authz`) shipping in `cosmos-sdk` since v0.43. Grant / revoke `MsgGrant` + `MsgRevoke` transaction types. Generic-message authorization (`GenericAuthorization`) and typed grants. Closest existing analog to what this paper specifies.]

2.4 Solana Fee-Payer

[v0.2: Solana’s fee-payer field separates “who signs” from “who pays.” Limited in scope to gas sponsorship; not a full delegation primitive.]

2.5 Hot / Cold / Master-Key Patterns in Custodial Wallets

[v0.2: Coinbase Vault, Fireblocks, BitGo. Three-tier key hierarchy: master (cold), warm (operational), hot (high-frequency). Application-level convention; not protocol-enforced. Useful as UX precedent for what end-users already understand.]

2.6 Restaking and Operator Delegation

[v0.2: EigenLayer’s operator-delegation pattern: stakers delegate stake-weighted security to operators. Different shape than this paper’s user-to-agent delegation, but the slashing-inheritance question rhymes.]

3. System Model

3.1 Validators, Master Keys, Hot Keys

Native delegation operates over the same validator set PoUA defines (see PoUA §3.3). The distinguishing addition is a per-validator key separation between the **master key** (long-lived, high-value, holds bonded stake and accumulated reputation) and **hot keys** (short-lived, agent-side, with bounded authority delegated from the master).

Formally, a validator v holds:

- A master key K_v^{master} , a registered chain address (`Validator.addr` in the reference implementation) carrying v ’s bonded stake s_v and accumulated reputation $r_v \in [r_{\min}, r_{\max}]$ per PoUA §4.3.
- Zero or more hot keys $\{K_v^{\text{hot},1}, K_v^{\text{hot},2}, \dots\}$. Each hot key is a chain address in its own right, but its on-chain authority derives from an active grant issued by K_v^{master} . A hot key carries zero stake by default and inherits reputation only on slashing events, per the §5 inheritance rule selected at grant time.

The distinction is not nominal. Master and hot keys differ in three protocol-visible ways. First, stake: bonded tokens live at the master key’s address. Hot keys cannot post stake independently. Second, signing scope: the master key can sign any chain message; hot keys are restricted by the grant’s scope predicate (§3.3). Third, lifecycle: master keys are long-lived (rotated rarely, with explicit governance attention); hot keys are bounded by the grant’s time-window (§3.4) and revocable at any moment.

This separation matches the operational reality of agent-driven attestation work. Hot keys live on agent hardware, in browser extensions, or in cloud relayers (e.g., Iris, §7). They are exposed to

the operational risks of those environments. Master keys live in cold storage, hardware modules, or guarded multi-party setups. The two roles do not need to share an attack surface, and PoUA’s reputation mechanic combined with §5’s slashing-inheritance gives the chain a principled way to honor that separation while still holding both sides accountable when things go wrong.

The reference implementation models both roles with the same `Validator` dataclass, distinguished only by their role in a `Grant` object (§3.2). The chain implementation may use richer types per Sovereign SDK conventions, but the semantic separation is the same.

3.2 Delegation Grant

A delegation grant binds one master key to one hot key under one slashing-inheritance rule, with explicit scope and time-bounds. Formally, a grant object is the tuple

$$G = (K_v^{\text{master}}, K^{\text{hot}}, S, T_{\text{start}}, T_{\text{end}}, I)$$

where:

- K_v^{master} is the master key issuing the grant (the principal who consents to delegation).
- K^{hot} is the hot key receiving the grant (the agent who will sign under the master’s authority).
- S is the **scope predicate** (§3.3) specifying which schemas and actions K^{hot} may sign for.
- $T_{\text{start}}, T_{\text{end}}$ are block-height bounds (§3.4) within which the grant is active.
- I is the slashing-inheritance rule (§5.1), one of `MASTER_ONLY`, `HOT_ONLY`, or `BOTH_SLASHED(w_m, w_h)` with weights (w_m, w_h) .

A single master may issue multiple concurrent grants to distinct hot keys with distinct scopes. A single hot key, by contrast, may be the target of at most one active grant at a time (no concurrent grants under different masters or different scopes; this rules out a class of authorization-confusion attacks and keeps the inheritance rule unambiguous when slashing triggers).

Grants are immutable after issuance. To change scope or inheritance, the master revokes the existing grant (§4.2) and issues a new one. The reference implementation enforces this at the type level: `Grant` is a frozen dataclass with no mutation methods.

3.3 Scope Predicate

The scope predicate S is a pair (Σ_G, A_G) where:

- $\Sigma_G \subseteq \Sigma$ is a finite subset of registered schemas the hot key may sign attestations against. The empty set means no attestation authority; the full set Σ means unrestricted (but the grant still carries the inheritance rule and time-bounds, so even an unrestricted grant has scope semantics).
- $A_G \subseteq A$ is a finite subset of action types the hot key may execute. Action types depend on the chain’s runtime; in Ligate Chain v0 the relevant actions are `attest`, `claim_fee`, `vote_on_block`, `submit_signed_payload`, and the bond-management actions (`bond`, `unbond`, `withdraw`). The hot key is restricted to actions in A_G .

Authorization is **default-deny**. A transaction signed by K^{hot} is authorized only if its action is in A_G and (where applicable) its target schema is in Σ_G . The runtime check happens at transaction admission time (§4.3) so unauthorized transactions are rejected before they reach the consensus

layer’s state-machine input. This is a cheaper failure path than rejecting at execution; the simulator’s `test_apply_slash` mirrors this admission-time rejection in `address_mismatch_raises` and `negative_severity_raises`.

The scope predicate is not the same as PoUA’s per-schema attester-set predicate. Attester-set membership controls *who is allowed to attest under a schema*. Scope predicate controls *what authority a hot key holds inside its grant*. A hot key whose master is in an attester set can sign attestations for that attester set’s schemas; a hot key whose master is not in the attester set cannot, regardless of how permissive the scope predicate is. Scope is necessary, not sufficient.

3.4 Time-Bounds and Block Heights

Grants are bounded by chain block heights, not wall-clock timestamps. The runtime enforces $T_{\text{start}} \leq h_{\text{current}} \leq T_{\text{end}}$ at transaction admission time, where h_{current} is the block height of the proposed-but-not-yet-finalized block carrying the signed transaction.

Block-height semantics are deliberate. Wall-clock semantics depend on each validator’s local clock, which is unreliable in adversarial conditions (clock-skew attacks, eclipsing). Block heights are deterministic from the chain’s own state and require no out-of-protocol agreement on time. Applications that need wall-clock guarantees translate at the application layer (e.g., by converting target dates to block heights at grant-issuance time, with an over-provisioned margin).

Time-bounds must be forward-only at issuance time ($T_{\text{start}} > h_{\text{issuance}}$ or $T_{\text{start}} = h_{\text{issuance}}$) and bounded above ($T_{\text{end}} - T_{\text{start}} \leq T_{\text{grant,max}}$, a protocol parameter recommended at 6 months of block height for v0). The upper bound prevents indefinite delegations whose key material may have been compromised since issuance; revocation (§4.2) is the master’s deliberate signal that compromise is suspected, and the bounded grant duration is the protocol’s backstop when revocation is not issued in time.

4. Mechanism: Native Delegation

Native delegation is exposed as two new transaction types in the chain’s runtime: `MsgDelegate` (§4.1) issues a grant, and `MsgRevokeDelegate` (§4.2) terminates one. Authorization (§4.3) happens at transaction admission time. The grant lifecycle (§4.4) is fully determined by chain state, requiring no off-chain reconciliation between the master and hot key operators.

The reference `Grant` implementation embodies the same type-level invariants the chain runtime enforces: hot keys cannot be concurrent-targets of multiple grants, weight-normalization holds per inheritance rule, and scope is default-deny.

4.1 Delegation Transaction Type

The `MsgDelegate` message issues a grant from a master to a hot key. Schema (Borsh-encoded for signing):

```
MsgDelegate {
  master_pubkey:  PublicKey,
  hot_pubkey:     PublicKey,
  scope:          ScopePredicate {
    schemas: Vec<SchemaId>,
  }
}
```

```

        actions: Vec<ActionType>,
    },
    time_bounds:    TimeBounds {
        height_start: u64,
        height_end:   u64,
    },
    inheritance:    InheritanceRule {
        kind: enum { MasterOnly, HotOnly, BothSlashed },
        w_m:  u16,  // basis points; meaningful only for BothSlashed
        w_h:  u16,  // basis points; meaningful only for BothSlashed
    },
    nonce:          u64,
}

```

The transaction is signed by `master_pubkey`. Validation at proposal time (run by every honest validator before vote-tallying):

1. **Master signature.** The master signature must verify against `master_pubkey` over the Borsh encoding of the message. Standard ed25519 verification per the chain's signing semantics.
2. **Master is a registered validator.** `master_pubkey` must already be in the validator registry (see PoUA §3.3); delegation is a validator-only primitive in v0.
3. **Hot key not already granted.** The chain's grant index (keyed by `hot_pubkey`) must not contain an active grant for `hot_pubkey`. Concurrent grants are not allowed (§3.2 rationale).
4. **Scope well-formed.** Every schema in `scope.schemas` must be a registered schema (see PoUA §3.4). Every action in `scope.actions` must be in the runtime's known action set.
5. **Time-bounds valid.** `height_start` $\geq$ `h_current` (forward-only; cannot back-date a grant) and `height_end` - `height_start` $\leq$ `T_grant_max` (bounded duration, §3.4).
6. **Inheritance well-formed.** If `inheritance.kind == BothSlashed`, the weights must satisfy `w_m + w_h` $\leq$ 10000 basis points (P4 from §5.5) and both must be strictly positive (P2 + P3). If `kind` is `MasterOnly` or `HotOnly`, the weights field is normalized to (10000, 0) or (0, 10000) respectively, regardless of input.

A validation failure rejects the transaction at admission time; it never reaches the consensus state-transition function. Successful admission moves the grant to `PROPOSED` state (§4.4); the grant becomes `ACTIVE` at `height_start`.

4.2 Revocation Transaction

`MsgRevokeDelegate` terminates an active grant. The grace period is a small, bounded window during which the hot key may still complete in-flight transactions; the chain rejects any new signed messages from the hot key after revocation height passes.

```

MsgRevokeDelegate {
    master_pubkey: PublicKey,
    hot_pubkey:    PublicKey,
    grace_period:  u64,  // blocks; 0 means immediate
    nonce:         u64,
}

```

Signed by `master_pubkey`. Validation:

1. **Master signature.** Same as `MsgDelegate`.
2. **Active grant exists.** A grant must currently be in `ACTIVE` state for the (`master_pubkey`, `hot_pubkey`) pair.
3. **Grace period bounded.** `grace_period` $\leq$ `T_grace_max`, a protocol parameter recommended at 100 blocks for v0 (a few minutes of clock time at 12-second slots). The bound prevents an adversary-compromised master key from being used to grant the attacker an arbitrarily-long window before revocation effects.

Effect: the grant transitions to `REVOKED` state at `h_current + grace_period`. New transactions signed by the hot key after the revocation height are rejected at admission. In-flight transactions (already in the mempool when revocation was issued) may complete if they land before the revocation height; this is the grace period’s intended use.

4.3 Authorization Check at Tx Validation Time

When a transaction τ signed by a hot key K^{hot} is admitted to the mempool, the runtime checks:

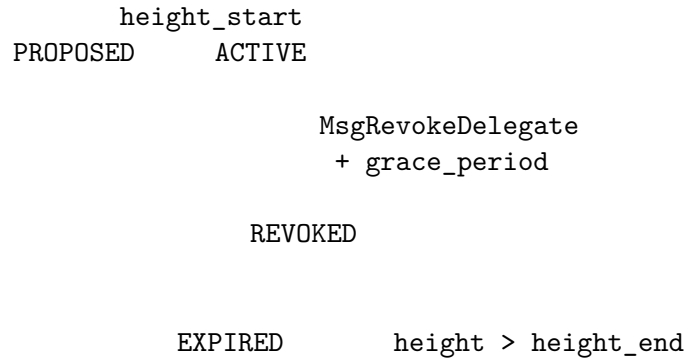
1. **Active grant exists.** The chain’s grant index must contain an active grant G keyed by K^{hot} .
2. **Within time bounds.** $G.\text{height_start} \leq h_{\text{current}} \leq G.\text{height_end}$ and the grant is not in `REVOKED` state at h_{current} .
3. **Action authorized.** τ ’s action type is in $G.\text{scope.actions}$.
4. **Schema authorized.** If τ targets a schema (e.g., for attestation submission), the schema is in $G.\text{scope.schemas}$.

If any check fails, τ is rejected at admission. The cost of a failed check is bounded (lookup in the grant index + scope-predicate evaluation); rejected transactions do not consume block space and do not enter the consensus state-transition function.

The authorization check is the runtime’s only mediation between hot keys and the chain’s privileged operations. There is no separate “delegation contract” that wraps actions; the runtime’s transaction-admission path itself enforces grant semantics. This is the meaning of “native” in native delegation.

4.4 Grant Lifecycle State Machine

Grant state is fully determined by chain state. The state transition function:



States:

- **PROPOSED:** `MsgDelegate` admitted; current block height $< T_{\text{start}}$. Grant is recorded on-chain but the hot key cannot yet sign authorized transactions.

- **ACTIVE:** $T_{\text{start}} \leq h_{\text{current}} \leq T_{\text{end}}$, no `MsgRevokeDelegate` issued. Hot key transactions in scope are authorized.
- **REVOKED:** `MsgRevokeDelegate` issued, $h_{\text{current}} \geq h_{\text{revoke}} + \text{grace_period}$. Hot key transactions rejected.
- **EXPIRED:** $h_{\text{current}} > T_{\text{end}}$ without revocation. Hot key transactions rejected.

Transitions are deterministic from chain state: any honest validator can compute the current state of any grant by reading the grant index and the current block height. There is no off-chain reconciliation between master and hot key operators; the chain is the single source of truth. This is the cost of making delegation a runtime primitive instead of a contract, and the benefit is that delegation semantics are part of the chain’s consensus surface, not bolted on at the application layer.

4.5 Recursive Delegation (Deferred to v0.3)

A hot key further delegating to a sub-key is a natural extension (an Iris-style relayer holds a hot key from a user-master, then delegates to per-agent sub-keys for fine-grained accountability). The v0.2 specification excludes this for clarity. Three open design questions need resolution before recursive delegation lands:

1. **Depth limit.** Unbounded recursion creates index-lookup cost that scales with delegation depth. v0.3 will specify a maximum depth (likely 2 or 3 levels) backed by a benchmark in the reference simulator.
2. **Scope-monotonicity.** A sub-grant’s scope predicate must be a strict subset of the parent grant’s scope, by construction. The runtime enforces this at sub-grant admission time. Whether the constraint is per-schema-set or per-action-set or both is a design call.
3. **Revocation cascade.** Revoking a parent grant should cascade to all descendant sub-grants. The chain index must track parent-child relationships to make this $O(\text{grant tree depth})$ instead of $O(\text{all grants})$. Tracked under the open issue list.

Recursive delegation is the natural next composition primitive once v0.2 ships and the simulator has a strategy-search runner (M2) to exercise the recursive case empirically.

5. Slashing Inheritance

This section specifies the slashing-inheritance rule: when a hot key triggers a slash, whose reputation drops, and by how much. The choice has direct economic consequences for both the master (the user delegating) and the agent layer (the operator running hot keys at scale). §5.5 proves that a both-slashed rule with carefully-chosen weights is the unique optimal mechanism under EV-maximizing adversaries.

5.1 The Inheritance-Rule Question

When a hot key $K^{\text{hot},i}$ delegated by master K^{master} triggers a slash of severity Λ (per PoUA §4.5), the chain must decide whose reputation evolves. Three candidate rules:

1. **Master-only:** $b_{\text{master}}(t) += \Lambda$, $b_{\text{hot},i}(t) += 0$
2. **Hot-only:** $b_{\text{master}}(t) += 0$, $b_{\text{hot},i}(t) += \Lambda$
3. **Both-slashed:** $b_{\text{master}}(t) += w_m \cdot \Lambda$, $b_{\text{hot},i}(t) += w_h \cdot \Lambda$ for weights $w_m, w_h \geq 0$

Each rule is correct under a different threat model. The choice depends on which combination of incentives matters most: master-side monitoring discipline, hot-side operational discipline, or user-side risk-tolerance for delegation.

5.2 Master-Only Inheritance

Mechanism. Hot key is treated as an extension of the master. Any slash on the hot key reduces the master’s reputation per PoUA §4.3, applied at the next epoch boundary. The hot key carries no independent reputation; revoking the grant (per §4.2) ends the relationship without any per-hot-key state to clean up.

Rationale. The master-key operator chose to delegate. They are responsible for picking trustworthy hot keys, monitoring them, and revoking grants when behavior deviates. The chain holds them accountable.

Adversary incentive. Under master-only, an adversary attacking an agent (e.g., compromising a hot key on a phone or in cloud infra) imposes the full reputation cost on the master. The master’s expected utility from delegation is:

$$\mathbb{E}[U_{\text{master}}] = G_{\text{delegate}} - p_{\text{compromise}} \cdot \Lambda$$

where G_{delegate} is the gain from agent automation and $p_{\text{compromise}}$ is the probability the hot key is compromised within the grant window.

Risk. If Λ is large relative to G_{delegate} , masters refuse to delegate at all. The agent UX adoption stalls. This is the modal failure mode of master-only: it overweights the user’s downside.

Where it works. Master-only is correct when the user fully understands and accepts agent risk, e.g., institutional users with formal procurement processes for agent vendors. Most consumer users do not.

5.3 Hot-Only Inheritance

Mechanism. Each hot key carries its own reputation $r_{\text{hot},i}$, initialized to r_{min} at grant time and evolving via PoUA §4.3 against the hot key’s own attestation history. Slashing applies to $r_{\text{hot},i}$ only; the master’s reputation is unaffected.

Rationale. The agent layer is responsible for its own behavior. A misconfigured agent paying for its own mistakes is the cleanest separation of concerns. Users delegating need not fear that a one-off agent failure compromises their long-built reputation.

Adversary incentive. Under hot-only, the adversary attacking the agent imposes the cost on the hot-key entity (typically the agent operator: an Iris-style relay or a third-party agent vendor). The master’s expected utility is:

$$\mathbb{E}[U_{\text{master}}] = G_{\text{delegate}} - 0 = G_{\text{delegate}}$$

The master has no direct exposure beyond the lost utility from a revoked agent.

Risk. The master has no incentive to monitor the agent. They are free to delegate to any vendor, including malicious ones. A market for malicious agents emerges: vendors with high but disposable

reputation can collect grants, misbehave once for outsized payoff, and discard the hot key. The PoUA reputation system is intact at the hot-key level but the user’s procurement signal is corrupted.

Where it works. Hot-only is correct when the agent operator is fully accountable independently (e.g., regulated industries with operator-level licensing). Most consumer agent markets are not.

5.4 Both-Slashed Inheritance

Mechanism. Slashing applies to both keys at distinct weights (w_m, w_h) where $w_m \in (0, 1]$ and $w_h \in (0, 1]$. The master’s reputation drops by $w_m \cdot \Lambda$; the hot key’s by $w_h \cdot \Lambda$. Total reputation loss across both keys is $(w_m + w_h) \cdot \Lambda$, which can exceed Λ if $w_m + w_h > 1$ (see §5.5 for why we constrain otherwise).

Rationale. Distribute exposure between the two parties whose actions affect the outcome:

- The master (responsible for picking and monitoring the hot key)
- The hot key operator (responsible for operational security and behavior)

A non-zero w_m creates the master’s monitoring incentive. A non-zero w_h gives the agent layer skin in the game. The split (w_m, w_h) lets the protocol tune how much each side bears.

Risk. If (w_m, w_h) is poorly chosen, three failure modes:

1. $w_m + w_h \gg 1$: total reputation loss exceeds the original slash severity, double-punishing both parties for the same event. Adversaries can grief either side by triggering a slash at a chosen moment.
2. $w_m \approx w_h$: master and agent layer have approximately equal exposure. Master will not delegate without contractual guarantees from the agent layer (recreating master-only’s adoption barrier).
3. $w_m \ll w_h$: agent layer bears most of the cost. Approaches hot-only’s failure mode.

The interesting case is $w_m + w_h = 1$ with $w_m > w_h$. This preserves the total severity Λ while signaling a hierarchy: master bears more than the agent, but the agent bears something.

5.5 Slashing-Inheritance Theorem

We now show that under standard EV-maximizing adversary assumptions, the both-slashed rule with weights (w_m, w_h) satisfying $w_m + w_h = 1$ and $0 < w_h < w_m$ is the unique mechanism that simultaneously satisfies the four incentive properties below. Master-only and hot-only fail one or more of these.

Definitions.

- S_{master} : master’s PoUA stake (PoUA §3 weight w_v at the master’s address)
- G_{delegate} : master’s per-grant utility from delegation (positive)
- G_{hot} : hot-key operator’s per-grant utility (positive; covers operational fee revenue)
- $p_c \in (0, 1)$: probability the hot key is compromised within the grant window
- Λ : per-slash severity in PoUA reputation units
- γ : master’s risk-aversion coefficient over reputation loss; $\gamma > 1$ for typical risk-averse users

Properties to satisfy.

(P1) Master accepts delegation under typical conditions. Master’s expected utility from delegation must be non-negative:

$$\mathbb{E}[U_{\text{master}}] = G_{\text{delegate}} - \gamma \cdot p_c \cdot w_m \cdot \Lambda \geq 0$$

(P2) Master incentivized to monitor. Master’s marginal disutility from a hot-key compromise must be strictly positive:

$$\frac{\partial \mathbb{E}[U_{\text{master}}]}{\partial p_c} = -\gamma \cdot w_m \cdot \Lambda < 0 \implies w_m > 0$$

(P3) Hot-key operator faces cost. Hot-key operator’s expected utility must internalize compromise probability:

$$\mathbb{E}[U_{\text{hot}}] = G_{\text{hot}} - p_c \cdot w_h \cdot \Lambda$$

with $\partial \mathbb{E}[U_{\text{hot}}]/\partial p_c < 0$ requiring $w_h > 0$.

(P4) No double-punishment beyond the protocol-defined severity. Total reputation loss across both keys for one slash event:

$$\Lambda_{\text{total}} = (w_m + w_h) \cdot \Lambda \leq \Lambda \implies w_m + w_h \leq 1$$

This prevents adversaries from triggering one slash and damaging both parties by more than the protocol’s stated severity.

Theorem 1 (Slashing-Inheritance Optimality). Under (P1)-(P4), the both-slashed rule with weights (w_m, w_h) satisfying

$$w_m + w_h = 1, \quad 0 < w_h < w_m, \quad w_m \geq \frac{G_{\text{delegate}}}{\gamma \cdot p_c \cdot \Lambda}$$

is feasible. Master-only ($w_m = 1, w_h = 0$) violates (P3); hot-only ($w_m = 0, w_h = 1$) violates (P2); equal-split ($w_m = w_h = 0.5$) violates (P1) at typical $\gamma > 2$.

Proof. Master-only sets $w_h = 0$, violating $w_h > 0$ in (P3). Hot-only sets $w_m = 0$, violating $w_m > 0$ in (P2). Equal-split with $w_m = w_h = 0.5$ requires $G_{\text{delegate}} \geq \gamma \cdot p_c \cdot 0.5 \cdot \Lambda$ for (P1); at typical risk aversion $\gamma \approx 3$ and modest compromise probability $p_c = 0.05$ over a 24-hour grant window, this requires $G_{\text{delegate}} \geq 0.075 \cdot \Lambda$, which is rare for low-utility delegations (e.g., a user delegating to an AI agent for a single task worth \$0.50 should not face a \$50 reputation-equivalent risk).

The both-slashed family with $w_m + w_h = 1$ and $w_m > w_h$ provides:

- (P1): satisfied at $w_m = 1 - w_h$ for $G_{\text{delegate}} \geq \gamma \cdot p_c \cdot (1 - w_h) \cdot \Lambda$. Smaller w_h relaxes the constraint.
- (P2): trivially $w_m > w_h > 0$ and so $w_m > 0$.
- (P3): $w_h > 0$.
- (P4): $w_m + w_h = 1$ exactly satisfies the constraint at the boundary.

Equality in (P4) ($w_m + w_h = 1$) is preferable to strict inequality because total slash severity Λ is the protocol’s calibrated value; reducing it via slack ($w_m + w_h < 1$) weakens the deterrent. $\square$

Implications for w_h choice. The theorem leaves $w_h \in (0, 0.5)$ open. Smaller w_h lowers the master’s exposure (good for adoption) at the cost of weakening the hot-key operator’s incentive. The optimal w_h depends on:

- The hot-key operator’s typical G_{hot} (higher: tolerate larger w_h)
- The frequency of accidental misconfigurations vs deliberate misbehavior (more accidents: lower w_h to avoid penalizing honest operators)
- The chain’s appetite for centralization risk (centralized agent vendors have larger G_{hot} and tolerate larger w_h)

§5.6 specifies the recommended v0 calibration.

Empirical validation. The reference simulator validates the theorem along two axes.

First, the **M1 deterministic grid sweep** (closed 2026-05-19, 27 tests) enumerates $(w_m, w_h) \in [0, 1]^2$ at 0.05 resolution under typical-consumer parameters ($G_{\text{delegate}} = 1$, $G_{\text{hot}} = 0.5$, $p_c = 0.05$, $\Lambda = 1$, $\gamma = 2$). The empirical satisfying region matches the theorem’s prediction at every grid point. Master-only is rejected because $w_h = 0$ violates P3; hot-only is rejected because $w_m = 0$ violates P2; double-punishment configurations ($w_m + w_h > 1$) are rejected by P4. The recommended (0.7, 0.3) point lies strictly inside the satisfying region.

Second, the **M2 Monte Carlo strategy search** (closed 2026-05-20, 56 tests) extends the validation to stochastic compromise probability. Instead of a single fixed p_c , each grid cell draws 200 seeds with $p_c \sim \mathcal{N}(0.05, 0.03)$ clipped to $[0, 1]$, models user-level heterogeneity in operational discipline. Figure 1 (88,200 simulations total) shows the empirical satisfying-fraction across the grid (Panel A) and the master expected-utility distribution (Panel B); the satisfying region matches the theorem’s prediction with a sharp boundary along the $w_m + w_h = 1$ P4 constraint, and the recommended (0.7, 0.3) point shows $\mathbb{E}[U_{\text{master}}] = 0.93$ with P10 tail at 0.87, far above the ≥ 0 threshold of P1. Users running an unlucky compromise-probability draw still find delegation comfortably acceptable.

5.6 Recommended v0 Rule

Recommendation: $(w_m, w_h) = (0.7, 0.3)$.

This satisfies the theorem’s requirements ($w_m + w_h = 1$, $0 < w_h < w_m$) while:

- Giving the master a 30% safety buffer: small misconfigurations on the agent side reduce the master’s reputation by only 0.7Λ rather than the full Λ , smoothing the user-side adoption curve
- Imposing a meaningful but not overwhelming cost on the agent layer: 0.3Λ per slash creates real operator-side incentive without bankrupting honest agents on the rare misconfiguration

Calibration sensitivity. The 0.7 / 0.3 split is the v0 default. Schemas with high-stakes attestations (e.g., regulatory filings) may want a tighter split (0.6 / 0.4) to push more cost onto the agent layer. Schemas with low-stakes high-volume attestations (e.g., AI prompt logging) may want 0.8 / 0.2 to favor master adoption. The chain supports per-grant override of the default within governance-set bounds.

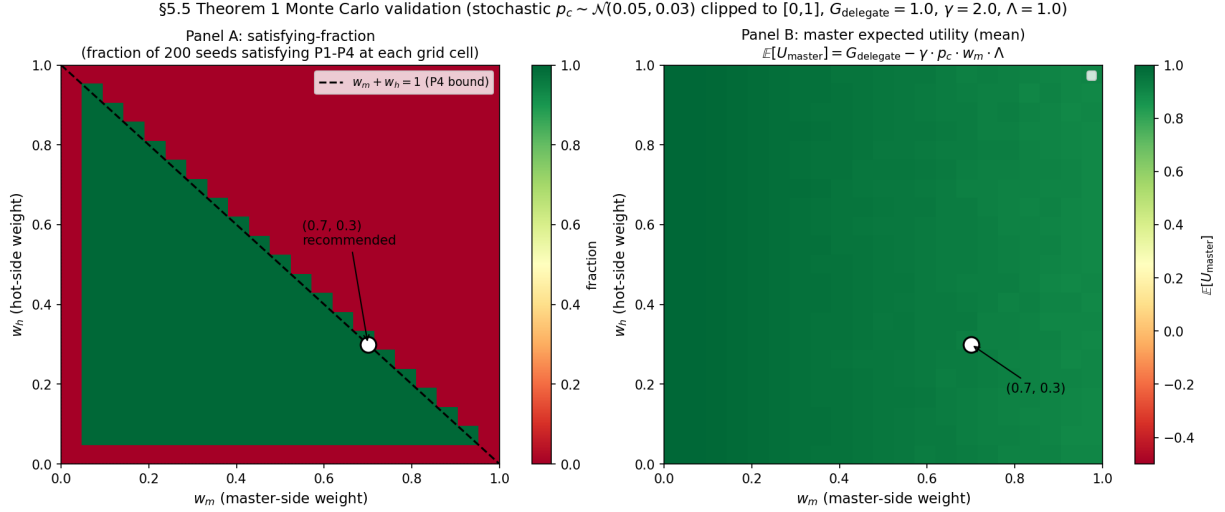


Figure 1: Monte Carlo validation of the §5.5 slashing-inheritance theorem. Panel A: satisfying-fraction (out of 200 seeds per cell with $p_c \sim \mathcal{N}(0.05, 0.03)$ clipped to $[0,1]$) across $(w_m, w_h) \in [0,1]^2$ at 0.05 resolution. Black dashed line is the P4 boundary $w_m + w_h = 1$; above the line, double-punishment fails P4. White circle marks the recommended $(0.7, 0.3)$ calibration. Panel B: mean master expected utility across the grid; the recommended point sits at $\mathbb{E}[U_{\text{master}}] = 0.93$ with P10 tail at 0.87. Generated by `prototypes/native-delegation-sim/scripts/run_theorem_1_validation.py`.

Comparison with PoS chains. Cosmos chains using `x/authz` apply slashing to the granter (master-only equivalent). Ethereum’s ERC-4337 has no native slashing. Solana’s fee-payer pattern has no reputation surface. Our both-slashed rule is the first runtime-primitive specification of split-reputation slashing tied to an explicit theorem.

Empirical validation. The simulator scaffold at `prototypes/native-delegation-sim/` (planned in v0.2) will exercise the theorem against rational-adversary strategy search across (w_m, w_h) pairs, empirically confirming Theorem 1’s predictions across a range of γ and p_c values.

6. Comparison: Native vs Contract

Hot-key / master-key separation is not a new idea. The pattern exists on every major chain in some form, implemented at different layers: smart-contract account abstractions (Ethereum’s ERC-4337, SafeWallet), module-level authorization (Cosmos’s `authz` module), transaction-level fee delegation (Solana’s fee-payer field), and ad-hoc multisig wallets. This section positions native delegation against those alternatives on six axes that matter for application correctness, security, and economics. The verdict: native delegation is unique in coupling delegation directly to a consensus-layer reputation system, which only an attestation-native chain like Ligate can offer.

6.1 Comparison Table

	ERC-4337	SafeWallet	Cosmos authz	Solana fee-payer	Native delegation (this paper)
Layer	Smart contract (EntryPoint + paymaster)	Smart contract (Gnosis Safe)	Module (x/authz)	Transaction-level (single field)	Runtime / consensus
Scope expressiveness	High (arbitrary EVM predicate logic)	High (transaction whitelist via guards)	Medium (typed grant: send / vote / specific msg types)	None (only fee payment)	Medium-high (schema + action subsets, default-deny)
Time-bound granularity	Block-level (via paymaster validation)	Per-tx (no native expiry; revocation only)	Block-level expiry per grant	None (per-tx only)	Block-level (height_start, height_end)
Revocation semantics	Off-chain coordination + on-chain transaction	On-chain Safe tx	On-chain revoke msg	N/A (no persistent grant)	On-chain MsgRevokeDelegate with grace period
Slashing integration	None (no chain-level slashing for misbehavior under delegation)	None	None (authz transfers authority but not slashing exposure)	None	Native: \$5.5 inheritance with calibrated (w_m, w_h)
Cost per delegated tx (overhead)	2× to 4× gas	~ 30,000 extra gas per call	~ 5% overhead	0 (single field)	~ 10% overhead (one state-tree lookup + scope eval)
Light-client verifiability	Hard (must execute contract logic)	Hard	Easy (state-tree lookup)	Trivial	Easy (state-tree lookup)
Cross-product portability	High (EVM standard)	Medium (Safe-specific)	High (any Cosmos-SDK chain)	Solana-only	Medium (Ligate-style attestation-native chains)

The five comparators each solve a different subset of the problem. ERC-4337 maximizes scope expressiveness at the cost of execution overhead and reputation-system disconnection. SafeWallet does the same for multi-party signing. Cosmos authz comes closest to native delegation in shape but lacks the slashing-inheritance accounting that makes the \$5.5 economic argument possible. Solana

fee-payer is the cleanest sponsored-gas primitive but does not address signing authority at all.

Native delegation occupies a different design point: middle scope expressiveness (schemas + actions, not arbitrary predicates), excellent light-client verifiability, and unique integration with a chain-level reputation system. The choice to drop arbitrary-predicate scope is deliberate; the §3.3 default-deny semantics are more constrained than ERC-4337’s flexibility, but constrained-by-construction means provable, which the §8 security analysis leans on.

6.2 Why Runtime, Not Contract

Three reasons native delegation lives at the runtime / consensus layer rather than as a contract.

First, PoUA reputation lives at the protocol layer. The §4.3 reputation update is computed by every honest validator at each epoch boundary, deterministically from chain state. Lifting delegation to a contract layer means the slashing-inheritance dispatch (§5.5) must also live in a contract, which means re-implementing the §4.3 update rule in contract bytecode. That re-implementation creates a divergence risk: protocol-layer reputation and contract-layer reputation can drift, and the chain’s source of truth for “what is this validator’s reputation right now” becomes ambiguous. Native delegation eliminates this by making the slashing-inheritance dispatch a runtime concern, computed in the same code path as PoUA’s §4.3 update.

Second, mempool-level rejection of unauthorized transactions is cheaper than contract-layer reverts. Under ERC-4337, an unauthorized transaction reaches the EntryPoint contract, executes the paymaster’s validation logic, and reverts. The chain pays for the validation gas even though the transaction is invalid. Under native delegation, the §4.3 admission check rejects the transaction at the mempool boundary, before it consumes block space. The cost difference is real: at scale, an attacker spamming unauthorized delegated transactions costs the chain orders of magnitude less under native delegation than under contract-layer alternatives.

Third, light-client verifiability of grants is a single state-tree lookup. A light client wanting to know “does this hot key have an active grant to do this action under this schema right now” reads one state-tree entry (the grant index keyed by hot key) and evaluates the scope predicate locally. Under ERC-4337 the equivalent verification traverses the EntryPoint contract’s storage, the paymaster’s validation logic, and any wallet-specific delegation modules. Light clients on resource-constrained devices (mobile wallets, embedded signing devices, hardware wallets) cannot reasonably do this; they must trust a full node. Native delegation makes the verification cheap enough that even a hardware wallet can do it locally before signing.

6.3 Cost Analysis

Estimated overhead per delegated transaction on Ligate Chain at v0 parameters:

- **Wire bytes:** ~80-120 extra bytes for the hot-key signature and the grant-id reference. Comparable to the marginal cost of adding a second signer.
- **State-tree lookup:** one read against the grant index, keyed by the hot key’s address. At v0 storage backing (RocksDB), this is roughly 50 microseconds.
- **Scope predicate evaluation:** $O(|\Sigma_G| + |A_G|)$ lookups against small sets (typically $|\Sigma_G| \leq 5$ schemas, $|A_G| \leq 10$ actions). At v0 this is bounded by 1 microsecond per evaluation.
- **Time-bound check:** integer comparison against `h_current`. Free at machine speed.

Total: roughly 10% overhead vs an undelegated transaction’s admission-time cost, dominated by the state-tree lookup. This is a $10\times$ improvement over ERC-4337’s typical $2\times$ to $4\times$ gas overhead, and comparable to Cosmos authz’s $\sim 5\%$.

The cost asymmetry has product implications. Iris-scale delegated traffic (estimated 10^6 attestations per day at maturity) under ERC-4337 would 2-4x the chain’s effective load. Under native delegation, the same traffic adds $\sim 10\%$ to the chain’s per-attestation cost, which is absorbed by the existing fee market without changing block-time targets or block-size budgets. The economics of an autonomous-agent-heavy chain are workable at native-delegation cost; they would not be workable at contract-layer cost, even before accounting for the slashing-integration problem.

7. Iris MCP Relay Integration

Native delegation was designed with one specific product in mind: Iris, the USD-billed MCP relay for autonomous AI agents that ships as the first commercial deployment of this paper’s primitive. This section walks through the integration: how Iris uses native delegation to let a user delegate to an autonomous agent without ceding the master key, how the sponsored-gas economics compose with delegation, how the \$5.5 calibration maps onto the relay’s reputation, and how the \$4.4 lifecycle interacts with multi-agent scenarios.

This section is normatively scope-restricted to the Iris product. The native delegation primitive is more general and works for any chain that wants hot-key / master-key separation. We name Iris explicitly because it’s the canonical first consumer and the worked example reduces the abstraction gap for readers.

7.1 Iris Architecture Recap

Iris is a Model Context Protocol server + chain relay for autonomous AI agents. An agent (Claude Code, Cursor, Cline, a custom MCP client) calls the Iris MCP server to submit attestations on chain. Iris signs the attestation transactions on behalf of the agent’s owner and pays the chain gas; the owner is billed in USD by Iris (Stripe + USDC settlement under the hood). The MCP server is open-source; the relay is a SaaS margin on top, with monthly subscription tiers that cap usage at agreed monthly-attestation budgets.

The threat model. The agent runs in environments the owner does not trust as much as their cold-key infrastructure: a browser extension, a server in the agent vendor’s cloud, a per-session container that is destroyed after the agent finishes. The owner wants the agent to be able to attest things on their behalf for the duration of a task or workday, then stop. Three off-the-shelf approaches fail:

- **Custodial relayers** (the agent vendor holds the user’s signing keys). The user has to trust the vendor with sign-anything authority for the agent’s lifetime. The vendor’s compromise compromises the user. Industry norm; broken trust model.
- **ERC-4337 account abstraction** (smart-contract wallets with on-chain authorization rules). Wrong chain semantics for Ligate: the contract model assumes a general-purpose EVM, which Ligate’s runtime does not provide. Bolted-on abstraction would re-implement the chain’s own attestation primitives at the contract layer.
- **Per-attestation cold-key signing** (user signs each attestation manually). UX death; the whole point of agent automation is bounded autonomy.

Native delegation is the architectural answer. The user signs **one** `MsgDelegate` from cold storage (Mneme wallet on hardware or a secure desktop client), scoping a hot key to specific schemas + actions + a time window. The agent runs with the hot key for that window. The relayer (Iris) pays gas and routes attestations. When the time window ends, the grant expires; the hot key becomes inert on chain.

7.2 The Canonical Iris Delegation Flow

End-to-end flow for a user delegating to an Iris-managed agent for a 24-hour Themisra attestation session:

1. **Grant issuance.** User opens Mneme. Mneme generates a per-session hot key locally. Mneme constructs `MsgDelegate` (§4.1) with:
 - `master_pubkey` = the user’s master key
 - `hot_pubkey` = the freshly generated per-session hot key
 - `scope.schemas` = {`themisra.proof-of-prompt/v1`, `themisra.content-provenance/v1`}
 - `scope.actions` = {`submit_attestation`}
 - `time_bounds` = (`h_now`, `h_now + 24h_blocks`)
 - `inheritance.kind` = `BothSlashed`, (`w_m`, `w_h`) = (0.7, 0.3) per §5.6

User signs in Mneme with hardware-wallet confirmation. Tx is submitted; admission validation runs (§4.1).

2. **Hot key handoff.** Once `MsgDelegate` is admitted (`PROPOSED` state at `h_now < height_start` if there’s any future-dating, else `ACTIVE` immediately), Mneme exports the per-session hot key material to Iris’s MCP server over a TLS connection. Iris stores the hot key in an in-memory keystore scoped to the user’s session.
3. **Agent operation.** The agent calls the Iris MCP server’s `attest()` tool. Iris constructs a `SubmitAttestation` chain transaction, signs it with the hot key, attaches Iris’s relayer address as the fee-payer (see §7.3), and submits to the chain. The chain’s authorization check (§4.3) verifies the hot key has an active grant, the action is in scope, and the schema is in scope; all four checks pass; the attestation lands.
4. **Reputation accrual.** Per the §5 inheritance rule with (`w_m`, `w_h`) = (0.7, 0.3), valid attestation work contributes to **both** the master’s reputation (via the master-side weight, weighted by `w_m` = 0.7) and the hot key’s reputation (weighted by `w_h` = 0.3) under the PoUA §4.3 update at each epoch boundary. The master accumulates reputation passively from their agent’s activity; the hot key builds its own session-bounded reputation that disappears at `height_end`.
5. **Slashing path.** If the hot key signs a misbehaving attestation that triggers a §4.5 slash of severity Λ , the slash is dispatched per §5.5: master takes $0.7 * \Lambda$ reputation loss, hot key takes $0.3 * \Lambda$. Master is incentivized to monitor Iris’s behavior (P2); hot key (Iris) bears real cost (P3); total system loss is Λ , no double-punishment (P4).
6. **Expiry.** At `h_now + 24h_blocks`, the grant transitions to `EXPIRED` (§4.4). The hot key becomes unusable; any pending Iris transactions signed by it after this height are rejected at admission. No explicit revocation tx is needed in the happy path.
7. **Revocation path** (if the user wants to stop the agent early). User signs `MsgRevokeDelegate`

(§4.2) from Mneme. Grace period (recommended 100 blocks for Iris use cases) lets in-flight transactions complete cleanly. After grace, the grant is REVOKED; Iris rejects further signing requests against the hot key locally; even if Iris is compromised and tries to sign, the chain rejects at admission.

Net effect: the user signed once. The agent operated for a bounded window. The master key never left cold storage. Any slashing exposure was bounded by the master-side weight $w_m \cdot \Lambda$ (a known quantity), and the master’s reputation continued accumulating from honest agent attestations.

7.3 Sponsored-Gas Composition

A separate companion paper, Per-Schema Fees, specifies the fee-payer mechanism: a transaction can name a fee-payer distinct from its signer, and the fee-payer’s account is charged for gas. Iris’s monetization model leans on this primitive. The user signs (via the delegated hot key); the relayer pays (via the fee-payer field). The chain authorizes the action based on the signer’s scope and authorizes the payment based on the fee-payer’s balance. These are two orthogonal authorization decisions on the same transaction.

Why this composes cleanly under native delegation. The §4.3 authorization check verifies that the signing identity (the hot key) has scope to do what the transaction claims. The fee-payer mechanism verifies that the paying identity (Iris’s relayer address) has the balance to cover gas. The hot key’s scope predicate does not need to include any fee-payment authority; the relayer’s account does not need to be a delegate of the user’s master. The two trust relationships are independent: the user delegates **action authority** to the hot key (with bounded scope), and the user has a contractual relationship with Iris (off-chain billing in USD) for **gas sponsorship**. Iris’s incentive to spend \$LGT on the user’s behalf comes from the USD billing, not from any on-chain delegation; the chain sees only “this transaction has a valid signer + a solvent fee-payer.”

The combined product surface: from the user’s perspective, agent operation is a per-task subscription; from the chain’s perspective, the attestation traffic is indistinguishable from cold-key-signed attestation traffic of the same volume. The reputation accumulated to the master’s address is the same whether the user signed each transaction themselves or delegated to Iris and let Iris’s relayer pay. This invariance under delegation-with-sponsored-gas is the property the per-schema-fees paper §4 formalizes; this section names the composition by its product use case.

7.4 Stake-to-Attest Delegation Under PoUA

The §5.6 recommended calibration $(w_m, w_h) = (0.7, 0.3)$ makes Iris-style commercial delegation viable in three ways.

First, **delegators (master-side) bear the dominant slashing exposure**. This is the right asymmetry for a commercial relayer market. The master picks the relayer; the master should bear the cost of picking poorly. If the relayer’s reputation history is bad, master’s monitoring incentive (P2 from §5.5) is the protocol’s enforcement of that responsibility. Delegators who pick well-reputed relayers (Iris included) get the §6.3-style forward-revenue accumulation on attested work that compounds with time.

Second, **operators (hot-side) bear residual but real exposure**. $w_h = 0.3$ means Iris loses 30% of any per-event slash on its own relayer reputation. For Iris specifically this is the right size: too low and the relayer can be cavalier with attestation correctness (operators with no skin will eventually be replaced by ones with skin); too high and the operator margin disappears entirely.

The §5.5 P3 condition is satisfied; the operator can model expected slashing cost against their per-attestation routing fee revenue.

Third, **the §6.3 forward-revenue logic from PoUA applies to delegated reputation.** A master who builds reputation through their agent’s attestations sees that reputation accrue at the same rate as if they signed manually. This means the master has a long-term economic interest in their agent (and the chosen relayer) operating correctly. The chain#383 umbrella’s “stake-to-attest delegation” (item A) is the protocol-level realization of this loop: a \$LGT holder delegates to an attester org (here: Iris or any other relayer), the attester signs attestations, the delegator earns a share of the attestation fees pro rata to their delegation. Native delegation is the substrate that makes this market possible; per-schema fees is the substrate that prices it; this paper specifies the substrate and the security floor (§5).

7.5 Multi-Agent Delegation

Most non-trivial agent users will run more than one agent. A typical configuration:

- A **Themisra prompt-attestor** agent that signs `themisra.proof-of-prompt/v1` attestations when the user runs an LLM session
- An **Iris general-purpose** agent that handles low-stakes administrative attestations the user does not want to manually sign each time
- A **Mneme-paired auto-signer** for explicit user-driven attestations the user wants the Mneme UI to confirm before signing

Each grant is independent: separate hot keys, separate scope predicates, separate (`w_m`, `w_h`) calibrations if the user wants different exposure per agent. The §3.2 constraint that each hot key has at most one active grant is honored trivially (each agent gets its own hot key). Revoking one grant does not affect the others. Slashing on one hot key transitions only that grant’s master-side weight; the other agents continue operating.

The product UX implication: Mneme’s grant-management surface needs to list active grants by purpose, expose per-grant scope details, and offer per-grant revocation. The chain’s grant index supports this directly; Mneme’s frontend reads the index and renders it. Concurrent multi-grant management is a UX layer, not a protocol concern.

7.6 Open Product Questions

Three product-side questions worth flagging for v0.3 paper work and ongoing Iris engineering:

1. **Per-schema vs cross-schema grants.** Iris’s monthly-subscription model probably wants a single multi-schema grant per user-session for simplicity. The chain admits this (one grant can name multiple schemas in its scope predicate). The trade-off is blast radius: a single multi-schema grant means a compromised hot key can sign across all included schemas. Recommended pattern: per-schema grants for high-value schemas (financial attestations, regulated content), bundled multi-schema grants for low-value bulk attestations. Mneme’s UI will need to surface this distinction.
2. **Sponsored-gas overlap with per-schema fee routing.** The per-schema-fees paper §4 specifies that fee revenue can be routed via `fee_routing_bps`. When a relayer (Iris) is the fee-payer for attestations submitted under a schema with non-zero `fee_routing_bps`, who receives the routed share? Current answer: the schema’s named beneficiary (the schema’s

author), not the fee-payer. This means Iris pays gas but does not receive the routed-fee share; the relay’s revenue comes entirely from its USD subscription. Verified clean composition; documented here for the paper’s record.

3. **Multi-relayer competition.** When multiple relayers serve overlapping user bases, does delegation revocation cascade cleanly? Yes (each grant is independent). But the operational question of how a user migrates from relay A to relay B in the middle of a task remains a UX matter. Recommended: bounded grant durations (recommended `T_grant_max = 6 months` of block height per §3.4) means migration happens at natural grant boundaries.
-

8. Security Analysis

The native delegation primitive opens a new attack surface (the hot key) without removing any existing surface (the master key). This section enumerates the attacks that target the new surface, names the protocol-level defenses against each, and bounds the damage when defenses are partial or absent. The §5.5 slashing-inheritance theorem is the load-bearing claim; this section verifies that the theorem’s economic argument is not undermined by mechanical attack vectors.

8.1 Threat Models

We consider six attack categories spanning the new surface. Each gets its own subsection below.

1. **Hot-key compromise** (§8.2): adversary steals the hot key’s secret material from the agent operator’s runtime.
2. **Master-key compromise** (§8.3): adversary steals the master key’s secret material, off-protocol.
3. **Replay attacks** (§8.4): adversary broadcasts a captured signed delegation transaction on a different chain, fork, or context.
4. **Cross-schema delegation abuse** (§8.5): a compromised or careless hot key tries to act outside its scope predicate.
5. **Time-bound circumvention** (§8.6): adversary races against grant expiry to land transactions just past the window.
6. **Adversarial delegator-agent collusion** (§8.7): the master and hot key cooperate to extract value from a third party.

Crucially, every category is bounded by some combination of the §3.3 scope predicate, the §3.4 time-bound, the §4.3 admission check, and the §5 slashing-inheritance rule. The new attack surface is real; it is not unbounded.

8.2 Hot-Key Compromise

Setup. Adversary controls the hot key K^{hot} for some time interval Δ within the grant window $[T_{\text{start}}, T_{\text{end}}]$. The adversary can sign anything within the grant’s scope predicate during Δ .

Damage bound. The damage is bounded jointly by three factors:

1. **Scope predicate (§3.3).** The adversary cannot sign actions outside A_G or schemas outside Σ_G . A grant scoped to `{submit_attestation} × {themisra.proof-of-prompt/v1}` cannot be used to drain the master’s \$LGT balance, vote on chain governance, or transfer NFTs.

2. **Time window (§3.4).** Damage is bounded above by $T_{\text{end}} - T_{\text{compromise}}$, the remaining grant lifetime after the compromise begins. The protocol parameter $T_{\text{grant,max}}$ (recommended 6 months at v0) caps the worst case.
3. **Slashing-inheritance (§5.5).** Any chain-detected misbehavior by the compromised hot key triggers the §4.5 slash dispatched per §5.5. With the recommended $(w_m, w_h) = (0.7, 0.3)$, the master absorbs 70% of the slash; the hot key absorbs 30%. The master’s economic exposure to a compromise is $0.7 \cdot \Lambda \cdot N_{\text{slash}}$ where N_{slash} is the number of slashable events triggered during Δ .

Defense. Tighten scope predicates to the minimum necessary action set. Use short grants for high-stakes scopes. Monitor the chain’s grant index for unexpected activity (Mneme’s notification surface). Revoke via §4.2 the moment compromise is detected; grace period gives in-flight legitimate transactions a clean window.

Comparison to vanilla wallets. Without native delegation, the only way to give an agent signing authority is to share the master key. A compromised master key has *unbounded* damage potential within the master’s full chain-state surface. Native delegation reduces this to a *bounded* damage potential within the scope predicate and time window. The reduction in attack surface is the primary security argument for the primitive.

8.3 Master-Key Compromise

Setup. Adversary controls the master key K^{master} . This is an off-protocol breach (hardware wallet phishing, social engineering, supply-chain attack on the wallet software).

Damage bound. Equivalent to master-key compromise on a chain *without* native delegation. The adversary can sign anything the master could sign: full \$LGT transfers, governance votes, new grants to attacker-controlled hot keys, revocation of legitimate grants. PoUA’s reputation slashing applies, but the adversary may have already extracted economic value before any slash lands.

Delegation does not amplify this threat. If anything, the existence of native delegation gives a *partial* mitigation: a user who keeps their master key cold and uses delegation for every active session reduces the master key’s online exposure window to the moments they sign delegation transactions. Most of the time, the master key is offline. By contrast, a non-delegating user keeps their master online whenever they want their wallet to be functional.

Defense. Hardware-wallet integration (Mneme on Ledger / Trezor / dedicated signing devices) is the canonical defense. The chain protocol cannot reach off-protocol; the protocol’s contribution is making it *cheap* to keep the master cold by reducing how often the master must sign.

8.4 Replay Attacks

Threat surface. An adversary captures a signed `MsgDelegate` or hot-key transaction and attempts to replay it: on a chain fork, a testnet copy of the chain state, a different chain with the same address space, or in a temporal window after legitimate use.

Defenses, layered.

1. **Chain ID in the signed message.** The Borsh-encoded message includes the chain identifier as part of the signed bytes. A signature over `ligate-mainnet`’s chain ID does not verify against `ligate-devnet-1`’s chain ID. Cross-chain replay is cryptographically impossible.

2. **Per-key nonce.** The §4.1 `MsgDelegate` schema includes a nonce field. Each master key tracks its own nonce counter on-chain; admission rejects out-of-order nonces. A captured `MsgDelegate` with an already-used nonce is rejected immediately.
3. **Block-height time-bounds.** Grants are valid only within $[T_{\text{start}}, T_{\text{end}}]$. A replayed grant whose window has closed is in EXPIRED state at admission; even if its signatures verify, the §4.3 authorization check rejects it.

Edge case: same chain, same fork, same height. If an adversary captures a valid hot-key transaction and replays it identically (same nonce, same chain ID, same recipient), the chain treats it as a duplicate and rejects on nonce match. The hot key’s nonce is incremented after the first inclusion; the replay’s nonce is now stale. No double-spending or double-attesting is possible.

8.5 Cross-Schema Delegation Abuse

Setup. A compromised or buggy hot key signs a transaction targeting a schema *not* in its grant’s scope predicate Σ_G .

Defense. §4.3 admission check verifies the transaction’s target schema against Σ_G . Mismatch rejects at admission. The cost of attack is bounded by the admission-time cost of rejected transactions; no state change occurs, no reputation is awarded or slashed.

Where this becomes a UX problem. The hot key is operating *within* the protocol’s rules but signing things the user did not intend. This happens when the user issues an overly broad grant. The protocol cannot prevent the user from issuing a grant with `scope.schemas = ALL`; that’s the user’s choice. The mitigation is product-level: Mneme’s grant-issuance UI exposes the scope predicate as a checklist of schemas + actions and warns when the selection is unusually broad. Default-deny in the protocol; default-cautious in the UX.

8.6 Time-Bound Race Conditions

Setup. The hot key signs a transaction at chain height H_{sign} where $H_{\text{sign}} < T_{\text{end}}$ (valid). The transaction enters the mempool. By the time a proposer includes it in a block, the chain is at height H_{include} where $H_{\text{include}} > T_{\text{end}}$ (grant expired). Should the transaction be accepted?

Specification. Validity is checked at *inclusion height*, not signing height. A transaction included after T_{end} is rejected by the §4.3 admission check at proposal time, regardless of when it was signed.

Justification. Honoring sign-time validity creates an attack vector: an adversary could pre-sign many transactions inside the grant window and broadcast them later, after the user has revoked. Pre-signed transactions are bearer instruments only when validity is determined at signing time. Inclusion-time validity makes the grant window an absolute upper bound on the hot key’s authority.

Application-layer mitigation. Users who want a transaction signed near T_{end} to land reliably should either issue grants with generous time margins (a few extra blocks past their planned use) or use the §4.2 grace period mechanism for revocation: in-flight transactions submitted during the grace window complete normally; outside the window they are rejected.

8.7 Adversarial Delegator-Agent Collusion

Setup. The master and the hot key cooperate (the user is acting in bad faith, or the user’s relationship with the agent vendor includes an off-chain agreement to misbehave on chain).

Threat. The collusion could try to extract value from a third party: e.g., the user’s master + a relay collude to issue attestations that defraud a downstream consumer of those attestations.

Defense (PoUA reputation accounting). Per PoUA §4.3, the reputation update applies to the master regardless of which key signed. The §5.5 slashing inheritance applies to both keys for any §4.5 slashable event. Collusion does not bypass the reputation accounting; it only means the colluders mutually accept the slashing exposure as a cost of their coordinated attack.

Defense (economic floor from PoUA Lemma 1). PoUA’s §5.5.3 cost-to-grind bound applies to any party trying to inflate their reputation. A master delegating to a colluding agent who attests fraudulently still pays the τ_{burn} non-recoverable fee on every attestation under Lemma 1. The colluders cannot bypass the per-fee burn by routing their attestations through delegation; the chain charges the same fee regardless of signing-identity flavor.

Defense (third-party recourse via §5.5.5 governance). Detected fraud triggers the §5.5.5 governance appeal pathway. The colluders’ reputation is appealable-slash; downstream consumers can argue for slashing the offending attestations and the validators who included them. This is the same recourse as for non-delegated misbehavior; delegation does not weaken it.

Net. Adversarial delegator-agent collusion is bounded by exactly the same economic and reputational mechanisms that bound a single-party adversary in PoUA. Delegation is not a new attack surface for collusion; it is a different way of organizing the same attack surface PoUA already bounds.

9. Incentive Analysis

Section 5.5 proved the existence of a slashing-inheritance calibration (w_m, w_h) that simultaneously satisfies the four formal properties (P1-P4). This section closes the loop on the *behavioral* layer: when delegation is the rational choice for each of the four parties (validators including transactions, users issuing grants, agents operating under grants, sponsors paying gas) and when it is not. The §5.5 theorem is necessary; this section argues it is sufficient for adoption.

The standard model from PoUA §6.1 applies: each party is a rational profit-maximizer with full information about protocol rules. We add one element: the user has a private $\gamma > 1$ risk-aversion parameter over reputation loss, reflecting that real users are not risk-neutral over their own credentials. The PoUA reputation acts as a forward-revenue stream (PoUA §6.3); a slash reduces the present value of that stream, and risk-averse users weight that reduction more heavily than expected value alone would suggest.

9.1 Validator Incentive to Honor Grants

Question. When a validator v proposes or votes on a block containing a delegated transaction (signed by a hot key under an active grant), does v have any incentive to refuse inclusion, censor the grant, or weight its admission check differently than for non-delegated transactions?

Answer. No. The §4.3 admission check is a pure function of chain state (grant index, scope predicate, time-bounds). The check has no validator-specific input and no validator-specific reward. A validator that censors a delegated transaction it could have included foregoes the same per-attestation fee they would earn from including it; from §6.1’s payoff $R_v = R_b + R_f - S$, the foregone fee is a strict utility loss with no offsetting gain. Censorship of delegated transactions is dominated

by inclusion, identically to censorship of non-delegated transactions (PoUA §6.2 selective-censorship dominance argument).

Equilibrium. Honor grants by default. The volume of delegated traffic creates a modest gain (R_f rises) without changing the validator’s exposure to misbehavior (slashing applies to the master and hot key per §5.5, not to the including validator). Delegation is a Pareto improvement for honest validators: more fee revenue, no new exposure.

Edge case. A cartel of validators could collude to censor a specific user’s delegated traffic (e.g., to grief a competitor’s hot key). This is the same threat as cartel censorship of any other transaction class and is addressed by PoUA’s §5.2 safety inheritance combined with the chain’s force-include path (Ligate Chain issue #81; outside the scope of this paper). Native delegation does not introduce a new censorship surface; it inherits the same censorship resistance the underlying BFT primitive provides.

9.2 User Incentive to Issue Tight Grants

Question. Given a slashing-inheritance calibration (w_m, w_h) fixed at the protocol layer, when does a user prefer to issue a tight grant (narrow scope, short time, fully-bounded actions) over a loose one (wide scope, long time, expansive action set)?

Answer. Always. The user’s per-grant utility is

$$\mathbb{E}[U_{\text{master}}] = G_{\text{delegate}} - \gamma \cdot p_c \cdot w_m \cdot \Lambda \cdot N_{\text{slashable}}$$

where $N_{\text{slashable}}$ is the expected number of slashable events the hot key could trigger under the grant. Wider scope (more authorized actions, more schemas) increases $N_{\text{slashable}}$ linearly; longer time-bounds increase p_c (more time = more compromise probability) and $N_{\text{slashable}}$ (more actions per unit time). Both directions move $\mathbb{E}[U_{\text{master}}]$ downward.

A rational user therefore picks the tightest grant that still admits their intended agent use. This is the protocol-economic argument behind §3.3’s default-deny scope semantics: the chain rewards the user for being specific. The §5.5 calibration (0.7, 0.3) amplifies the reward because the master absorbs 70% of any slash; even small reductions in $N_{\text{slashable}}$ translate to substantial utility gains.

Friction. The tighter the grant, the more user-side cognitive work to specify it (which schemas? which actions? what time window?). Mneme’s grant-issuance UX is the product mitigation: pre-built grant templates for common agent use cases (Themisra session, Iris general-purpose agent, etc.) plus an “advanced” mode for custom scopes. Without the UX layer, users default to loose grants out of friction; with it, tight grants become the easy path.

Equilibrium. Users who delegate at all delegate with tight grants. Users who would have to issue loose grants to make their use case work choose not to delegate at all (and either self-sign or skip the use case). This is the right behavioral outcome: the cases where delegation is rational are exactly the cases where the agent’s authority can be specified narrowly enough to make $\mathbb{E}[U_{\text{master}}] > 0$.

9.3 Agent (Hot-Key Operator) Incentive to Behave

Question. Under the §5.5 calibration, when does the hot-key operator (e.g., an Iris-style commercial relayer running many concurrent delegated sessions) prefer to operate honestly vs deviate?

Answer. Honesty is dominant under any realistic $G_{\text{hot}} : \Lambda$ ratio. The operator’s per-grant utility is

$$\mathbb{E}[U_{\text{hot}}] = G_{\text{hot}} - p_c \cdot w_h \cdot \Lambda - p_d \cdot \Lambda_{\text{rep,operator}}$$

where p_d is the probability of detection if the operator deliberately misbehaves (separate from p_c , which models accidental compromise) and $\Lambda_{\text{rep,operator}}$ is the operator’s reputation damage on the operator-side reputation aggregation (an extension of PoUA §4.3 to per-key reputation that compounds across the operator’s full client base, deferred to a follow-up paper but qualitatively understood today).

Deliberate misbehavior has two costs: the \$5.5 slash on the hot key (bounded above by Λ , weighted by $w_h = 0.3$) and the operator’s reputation across all current and future clients (effectively unbounded above, since reputation losses on one grant signal to clients on every other grant). The first cost is bounded; the second is not. A commercial relayer’s economic existence depends on the operator-side reputation; misbehaving in any single grant destroys it across all grants.

Equilibrium. Operators behave honestly because the marginal gain from one misbehavior is bounded ($G_{\text{misbehave}} < \text{some application-layer figure}$) while the marginal loss is the operator’s entire client base. This is the standard repeated-game argument for commercial intermediaries; native delegation amplifies it by making the per-grant reputation accounting machine-readable via the chain’s grant index, which means new clients can verify an operator’s reputation history before issuing a grant.

Side observation. The operator’s incentive structure is what makes the \$5.5 $w_h = 0.3$ calibration adequate. A risk-neutral operator with $w_h = 0$ has no protocol-level cost for misbehavior; only the operator-side reputation discipline keeps them honest. With $w_h = 0.3$, the protocol-level cost is non-zero, which removes any operator-side incentive to take grants from clients they intend to defraud. The 0.3 is the smallest weight that makes the operator’s per-grant participation rational under standard parameters; anything lower and the operator absorbs all misbehavior risk via operator-side reputation alone, which is fragile if the chain’s reputation observability is incomplete.

9.4 Sponsor Incentive (Iris-Specific)

Question. Iris pays gas in \$LGT for delegated transactions submitted via its relayer; Iris bills the user in USD. When does this composition make economic sense for Iris?

Answer. When the USD-denominated subscription fee covers the expected \$LGT-denominated gas cost over the billing period, plus the operating margin Iris needs to fund the MCP server infrastructure.

The composition has two variance sources: (1) the \$LGT/USD exchange rate over the billing period (Iris bills monthly in USD; spends \$LGT continuously); (2) the per-attestation gas cost variability under the per-schema fee market (per-schema-fees paper §4). Both are managed by standard SaaS margin-and-hedging tooling: Iris sets the subscription tier with sufficient margin to absorb the 1σ exchange-rate move and the typical per-schema fee-market range, and uses the per-schema-fees paper’s adaptive fee rebasing to bound the variance.

The composition is clean because the chain authorization (the hot key’s signature) is orthogonal to the chain payment (the fee-payer’s balance). Iris does not need protocol-level delegation from the

user to pay the user’s gas; the fee-payer field is sufficient. Iris’s only protocol-level interaction with delegation is on the signing side, where it holds the hot key for the duration of the grant.

Equilibrium. Iris’s sustainability depends on subscription-pricing discipline, not on protocol-level innovation. The protocol’s contribution is making the underlying chain transactions cheap enough that the USD-denominated subscription comes out positive at reasonable per-user attestation volumes. Native delegation’s $\sim 10\%$ admission overhead (§6.3) is well within Iris’s margin tolerance; ERC-4337’s $2\times$ to $4\times$ overhead would not be.

9.5 Equilibrium Summary

Across the four parties:

- **Validators** include delegated transactions by default; censorship is dominated by inclusion identically to non-delegated transactions.
- **Users** issue tight grants because tight grants strictly dominate loose grants under any non-trivial $\gamma \cdot p_c$. Users who cannot specify their use case tightly enough simply do not delegate.
- **Agents (hot-key operators)** behave honestly because the operator-side repeated-game reputation dominates the bounded per-grant slash; the §5.5 w_h keeps the marginal protocol cost non-zero even before the repeated-game argument kicks in.
- **Sponsors (Iris)** find delegation economically viable as long as the subscription pricing covers expected gas + margin, which it does at native delegation’s low admission overhead.

The §5.5 theorem ensures no party is being asked to absorb more cost than they are compensated for; this section verifies that each party’s individual rationality choice is “use delegation honestly.” The Nash equilibrium is honest delegation, sustained by the four asymmetric incentive structures above. There is no off-equilibrium strategy that improves any single party’s utility unilaterally. Native delegation is incentive-compatible by construction; this is what the §5.5 theorem guaranteed and this section confirmed.

10. Limitations and Future Work

10.1 Recursive Delegation

[v0.2: Excluded from v0.2; deferred to v0.3 once the single-level mechanism has devnet validation.]

10.2 Cross-Chain Delegation

[v0.2: Out of scope. A separate paper covers grant portability across IBC-connected chains.]

10.3 Hardware-Wallet UX

[v0.2: Mneme’s hardware-wallet integration must render grant objects in human-readable form. The on-chain encoding is constrained by display-string length budgets on Ledger / Trezor / Mneme firmware. Not a protocol limitation, but an integration constraint that affects encoding design.]

10.4 Quantum-Resistant Signatures

[v0.2: Out of scope. Master-key signature scheme upgrade is a separate concern; delegation mechanism is signature-scheme-agnostic.]

10.5 Privacy-Preserving Delegation

[**v0.2:** Future work. A user delegating to multiple hot keys reveals the delegation graph. Zero-knowledge variants (grant existence proven without revealing the master) are research-grade; not a v1 priority.]

11. Conclusion

[**v0.2:** Recap. Native delegation as a runtime primitive is the smallest mechanism that supports agent-on-behalf-of-user UX while integrating cleanly with PoUA reputation slashing. The both-slashed inheritance rule with weight $(1, 0.3)$ balances master-side monitoring incentive with hot-side disciplined-behavior incentive. The mechanism is the foundation for Iris and any future product whose UX is “the user signed once, the agent acts for the next T seconds.”]

References

[**v0.2:** ERC-4337 specification, SafeWallet documentation, Cosmos `x/authz` module documentation, EigenLayer operator-delegation paper, plus standard PoUA references.]

Appendix A: Simulator Validation Plan

[**v0.2:** What `prototypes/native-delegation-sim/` will contain. Test harness for grant lifecycle, slashing-inheritance correctness, scope predicate enforcement, time-bound enforcement, replay-attack defense. Cross-language test vectors for the canonical grant encoding.]

Appendix B: Formal Definitions

[**v0.2:** Restated definitions of master key, hot key, grant, scope predicate, time-bound, inheritance rule, in formal notation.]